

Exam History of Science

Date: Wednesday July 3rd, 2024

Lecturer: dr. D.J. Beckers

Time: 2h 15min

Course code: FEW X 400318

In general:

- (1) Answer in English sentences. Spelling and grammar should be correct either by English or by US standards. Just a few buzzwords never constitute an answer. Always explain yourself. Use appropriate examples to illustrate your answer.
- (2) This exam consists of 8 A-questions, 12 B-questions, 6 C-questions and a NO BONUS question. The A-questions are about the lectures; the B-questions refer to information from the book by Campbell-Kelly. The C-questions require you to combine the information from the book and lectures in your answer.
- (3) You pick four questions: 1 A-question, 2 B-questions and 1 C-question. Each question is worth 2 points. If you answer more A-questions, only the first will be considered. Likewise for the B- and C-questions.
- (4) If you're *not* entitled to the bonus point, you additionally do the NO BONUS question. This question is worth one point.

A-Questions:

1. Prejudice

During the lecture, examples were provided to show how history might work to clarify prejudice. Discuss two specific examples of prejudice that may be brought to light by history.

2. Creating AI

Discuss a history of 1950s cybernetics. In what sense was cybernetics a precursor to present-day AI?

3. Sound of software

Late 1950s electronic computers had speakers attached. Why were speakers added to these machines and what was new about them? In your essay, mention specific computers.

4. computer and banking

To what extent were banks instrumental in 1960s/70s development of real time computer systems?

5. Proof by computer

In the 1960s rose the idea to make computers check mathematical proofs. Discuss the main protagonists in that story and explain their (lack of) success.

6. Game culture

In the lecture the history of the rise of game culture was connected to the history of AI. Which were the different relationships between 1950 to 1980?

7. Histories of AI

To what extent does a history of AI reflect present-day AI?

8. Digital culture

To what extent is the phrase "digital culture" applicable to our present-day culture? Provide examples where it unquestionably is, and where it is mere accidental (a hype?) to use the phrase.

B-Questions:

1. First computer

Campbell-Kelly claims that it is difficult to point to The First Computer. Explain his line of thought. In your story, mention some of the machines that could be considered as first computers.

2. UNIVAC and IBM

Campbell-Kelly discusses the entrance of IBM in the computer industry as a reaction to UNIVAC. Discuss his line of thought.

3. RPG

Campbell-Kelly suggests that RPG was one of the ways in which IBM smoothened the path from punched card technology to electronic computing. Explain his line of thought.

4. Real time

Campbell-Kelly puts Jay Forrester at the core of development of real time computing. Discuss two reasons why this is fitting. Who is the second in line for this particular development?

5. Programming

Programming was called “a twofold series of translations” by Campbell-Kelly. Explain the line of thought. In your essay, indicate the initial (that is: 1950s) tools that were developed and / or used in these translations.

6. Computer utility

Discuss the 1960s “computer utility” movement. Provide at least two reasons for its' (initial) success. In your story, mention at least one of the people involved.

7. Steward Brand

What roles does Campbell-Kelly attribute to Steward Brand in his history of computing?

8. Broadening the appeal

The chapter in which Campbell-Kelly provides a history of the 1980s rise of personal computing is titled “Broadening the appeal”. Discuss two explanations for this chapter title.

9. BBN

Campbell-Kelly discusses the role of BBN in the rise of network technology. Explain the role of this company in the . In your essay, mention at least two

10. Globalization

Campbell-Kelly uses Apple, Microsoft and IBM as exemplar for three ways in which globalization in the computer industry happened. Explain his line of thought.

11. Web 2.0

To what extent was Web 2.0 a novelty?

12. Computing and environment

Campbell-Kelly discusses several ways in which computing has been disastrous to our environment. Elaborate on two of his examples.

C-Questions:

1. I think, therefore I am

Discuss two of the 1960s/70s shifts in thinking about what makes us human (as opposed to machines or animals), in the light of developments in computing / AI. In your essay, use the concept of digital culture.

2. SAP

The German company SAP was discussed as an example of a successful tech company outside the rising Silicon Valley. To what extent does this story contrast the story in the book by Campbell-Kelly?

3. Time sharing

Discuss the rise of time sharing systems as a prerequisite for computing. Use of the concept of agendas in your essay.

4. Internet

The Internet of the late 20th century was discussed from different points of view in the book and the lectures. Elaborate on two of the main differences.

5. Defining technology

Discuss two 1960s examples of computing as a defining technology (in Bolter's sense).

6. Timnit Gebru

Discuss two of the main problems with AI which Timnit Gebru (a.o.) address in their paper on stochastic parrots. In your essay use the concept of agendas.

NO BONUS

To what extent was the Dartmouth 1956 conference relevant to the history of AI.